Machine Learning II

Hands-on Project Units 3 & 4

Submission

- You may carry the project out in groups of four students.
- Must upload the deliverables to Moodle before the deadline(s).

Notes:

- **A single submission per group**.
- **Submit only via Moodle**; not, for example, via email.
- **Do not use Moodle comments** to deliver any sort of important information (e.g., group members). Include all relevant information in the deliverables.

Deliverables

1. A PDF file with a lab report containing the details of the work (maximum 10 pages). Must contain the names of group members.
2. A python notebook with the code used to carry out the analysis. The notebook must run, out of the box, on Google Colab. The notebook must be included in the deliverable, not shared via Google Colab.
3. For the final delivery, in addition:
   - the test set completed with predictions;
   - a brief deck of slides summarizing the project.

Notes

- Submit all files within a single zip file.
- If the size of the video exceeds the maximum size allowed in Moodle, links to services such as One Drive (1 TB available for UPM students), Google Drive o Dropbox are allowed. Make sure that files do not exceed 200 MB.

Phases

The project will be delivered in **two phases**, followed by a defense. Regarding the first phase:

- train and evaluate naive Bayes and random forest
- corresponds to 10% of the project's grade (no minimum grade, and it is therefore not strictly necessary);
- deadline: 05/05/2025 at midnight.

The second phase corresponds to project completion, submitting the final and complete versions of all deliverables:

- You will be provided a test set, without labels, and you will fill-in those labels and upload the updated test set predictions to Moodle. Part of the evaluation grade depends on the accuracy on the test set. Failing to provide the test set in the adequate format will be considered as having baseline accuracy.
- Corersponds to 40% of the project's grade.
- deadline: 24/05/2025 at midnight.

The defense will take place during the lecture time on 26/05/2025. Corersponds to 50% of the project's grade and it's mandatory; students who fail to attend the defense will fail the project. Some details:

- Each groups members will be evaluated.
- Any member can be asked about anything that's written in the report or implemented in the Collab.
- You may be asked to change code or implement something on the spot.

What to do?

Train classifiers

You will fit classifiers supplied training data and then evaluate it on given validation data. You will need to consider each of the following methods:

- Naive Bayes
- Discriminant Analysis
- Bagging
- Random Forest
- AdaBoost
- Gradient Boosting
- Stacking

You will seek to make each classifier as accurate as possible. You will report 5-fold cross-validation estimates of accuracy. Learning the models includes trying and choosing among different parameters, preprocessing steps and so on.

Analyze feature importance

Report and discuss permutation feature importance measures for the best model.

Data

The training set is contained in the train.csv file and the validation set in the test.csv file. Both files have 51 columns, with the last one, named 'class', corresponding to the target variable.

Report structure

The structure of the report is as follows:

1. Cover page. Include a cover page with title, authors, email, course, and date.
2. Introduction. Briefly explain the problem, the data, your overall approach (e.g., whether you used extensive hyper-parameter tuning or a more manual search) and give overview of your results.
3. Method. Indicate your strategy and reasoning for fitting each of the mandatory classifiers, justifying the choice of different parameters and configurations; discuss why certain options might make sense and while others might not and you decide not to attempt them. **Describe at least two configurations or preprocessing steps that you attempted but discarded because they did not improve performance**.
4. Results. Provide validation set accuracy results for each classifier. Provide any additional plots, model coefficients, model size descriptors (e.g., number of base models), training set results, predicted class probabilities, etc. that will support the discussion. Provide feature importance results for one of the ensemble methods.
5. Discussion: Discuss the obtain results and hypothesize about the possible limitations of the different classifiers together with your choices of parameters, the preprocessing steps, etc. Describe how the observed results, and the hypotheses regarding their performance, guide your subsequent attempts of fitting more accurate models, in an iterative approach. Also provide a detailed justification for models that performed poorly.

6. Conclusion. Provide conclusions regarding the project and your results. Discuss possible additional steps to look for improvements.
7. References (optional).

Results summary

Important: The Results section must begin with a table summarizing considered learners and their best results. There will need be a row for each scikit-learn estimator class, and a column for the best non-default parameter settings for that class as well as for the best training set and 5-fold CV accuracy obtained with those settings. For example:

Classifier	Best parameters	Train accuracy	5-fold CV accuracy
LinearDiscriminantAnalysis	shrinkage='auto'	0.85	0.75
RandomForestClassifier	n_estimators=100	0.85	0.75
AdaBoostClassifier	n_estimators=100	0.85	0.75
...	...	...	...

Grading

Achieving a grade of 5 requires:

- Delivering all that is required, as specified above;
- *Decent* accuracy with many or all of the different models, as well as sensible feature importance results;
- Absence of major conceptual or formal mistakes.

The rest of the grade will come from

a) getting high model accuracy;
b) making visible, in the report, that you understand the material.

In a sense, both a) and b) are somewhat open-ended as you have freedom as to what do you discuss and attempt as well as how hard you try to improve accuracy. The goal is not necessarily to try everything possible, but to show evidence of understanding the material by justifying your reasoning and learning strategy. Thus, while you may use hyperparameter tuning, it is very much valid to try fewer parameters, manually, and justify that choice of parameters. Ideally, you will discuss and relate the performance of the different models.

Additional criteria include:

- The quality and clarity of the explanations, justifications, the writing and the presentation.
- Use of plots and other simple aids that help understanding the report.
- Good scikit-learn coding practices, such as using Pipelines.
- Extra content: e.g., it can be relevant to fit a highly accuracy model that is not among the mandatory ones, and discuss its result, relating it to the performance of the other models.

Important: There will be penalties for not following submission instructions, such as exceeding the page limit, not providing the code, etc. Additional recommendations:

- Not everything that you do needs to appear in the report; keep the report concise and to the point. I can check the notebook for all relevant details.
- Avoid plain copy-paste of code and output in the report; present the results in a clear and concise way.
- Watch out with figures: for example, ensure the axis' scales are discernible.
- Watch out for failures of the presentation, such as audio not working during parts of the video.
- I encourage discussion of the achieved results, the motivation for trying some options and not others, and the reasoning behind the choices made, etc.